

## Performance Testing

|                      |                                                              |
|----------------------|--------------------------------------------------------------|
| <b>Date</b>          | 1 July,2026                                                  |
| <b>Team ID</b>       |                                                              |
| <b>Project Name</b>  | OptiCrop – Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 5 Marks                                                      |

### Step 1: Testing Overview

| Field                    | Details                                               |
|--------------------------|-------------------------------------------------------|
| <b>Testing Tool Used</b> | Apache JMeter, Postman                                |
| <b>Type of Testing</b>   | Load Testing, Stress Testing                          |
| <b>Target Module</b>     | Flask Prediction API (/predict), Crop Prediction Form |
| <b>Test Environment</b>  | Windows 11, Python 3.x, Flask                         |
| <b>Test Date</b>         | 1 July, 2026                                          |

### Step 2: Test Scenarios

| S.No | Test Scenario / Description                 | No. of Virtual Users | Duration (sec) | Expected Outcome                                |
|------|---------------------------------------------|----------------------|----------------|-------------------------------------------------|
| 1    | Single user submits crop prediction request | 1                    | 30             | Crop recommendation generated within 2 seconds  |
| 2    | Continuous crop prediction requests         | 100                  | 120            | Stable response time without failures           |
| 3    | Peak load testing                           | 200                  | 180            | Application remains responsive under heavy load |

### Step 3: Performance Test Results

| S.No | Metric                | Target Value | Actual Value | Status | Remarks                 |
|------|-----------------------|--------------|--------------|--------|-------------------------|
| 1    | Average Response Time | < 2 sec      | 1.31 sec     | Pass   | Fast prediction         |
| 2    | Maximum Response Time | < 5 sec      | 3.18 sec     | Pass   | Within acceptable limit |
| 3    | Throughput            | > 40 req/sec | 50 req/sec   | Pass   | Good request handling   |
| 4    | Error Rate            | < 1%         | 0%           | Pass   | No request failures     |
| 5    | CPU Utilization       | < 80%        | 60%          | Pass   | Efficient CPU usage     |
| 6    | Memory Utilization    | < 80%        | 55%          | Pass   | Stable memory usage     |

## Step 4: Observations & Analysis

- The OptiCrop application responded quickly under normal and peak workloads.
- The Random Forest model generated crop predictions within the expected response time.
- No request failures or significant delays were observed during testing.
- CPU and memory utilization remained within acceptable limits.
- The application demonstrated stable and reliable performance for multiple simultaneous users.

## Step 5: Screenshots / Evidence

